https://github.com/AnuragHarapanahalli/cnl-sem5/blob/main/2.py

```
binary = input("Enter a 4-bit binary number: ")

if len(binary) == 4 and binary[0] != '0'and all(bit in "01" for bit in binary):

  m = len(binary)

  r = 0
  while (2 ** r) < (m + r + 1):
    r += 1

  n = m + r

  hamming = ['0'] * (n + 1)

  j = 0
  for i in range(1, n + 1):
    if i & (i - 1) == 0:
      continue
    hamming[i] = binary[j]
    j += 1

  for i in range(r):
    p = 2 ** i
    parity = 0

    for j in range(1, n + 1):
```

```
            if j & p:
                parity ^= int(hamming[j])

        hamming[p] = str(parity)

    print("Hamming Code:", "".join(hamming[1:]))

else:
    print("Not valid")
```

